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Week 05: Monday, AST 5011: Astrophysical Systems

Explosions, Variable Stars, and Compact Remnants

Michael Coughlin cough052@umn.edu

With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

Core Collapse and Nucleosynthesis

In Lecture 07 we traced the nuclear energy sources that power stars, from the pp chains and CNO cycles that sustain hydrogen burning, through helium burning (triple-alpha and $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$), and on through the advanced burning stages — carbon, neon, oxygen, and silicon — each proceeding at higher temperatures and on progressively shorter timescales. We saw that the binding energy per nucleon peaks at ^{56}Fe , so fusion beyond the iron group is endothermic: it absorbs energy rather than releasing it. We also saw that quiescent silicon burning produces ^{56}Fe , while rapid burning (as in a core-collapse supernova) produces ^{56}Ni , which then decays to ^{56}Fe — a distinction that will matter when we model supernova light curves later in this lecture.

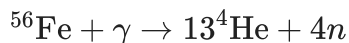
You have built an iron-rich core in your massive star — but what leads to core collapse?

The Start of Core Collapse

1. Photodisintegration

A nuclear process in which a nucleus absorbs a high-energy gamma ray, enters an excited state, and immediately decays by emitting subatomic particles: (γ, n) , (γ, p) , (γ, α) .

The most important reaction is the photodisintegration of iron:



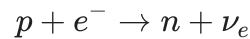
This process is endothermic — it absorbs about 124 MeV per iron nucleus, removing thermal energy and pressure support from the core. At $T \gtrsim 5 \times 10^9$ K, even the resulting alpha particles are destroyed: $^4\text{He} + \gamma \rightarrow 2p + 2n$, absorbing another ~ 29 MeV each. In effect, photodisintegration undoes all the nuclear fusion that powered the star through its life — the binding energy we carefully tracked in Lecture 07 is returned to the radiation field at the worst possible time.

2. Loss of thermal pressure support

As energy is absorbed by photodisintegration, the core can no longer maintain pressure balance. The effective adiabatic index γ_{ad} drops below the critical value of 4/3 needed for stability. Recall from Lecture 07 that thermal regulation normally works because nuclear energy generation is steeply temperature-sensitive ($\varepsilon \propto T^\nu$ with large ν), providing negative feedback. Once fusion ceases and photodisintegration drains energy, this self-regulating thermostat fails.

3. Electron capture (inverse beta decay)

Increasing density drives electron capture onto protons, reducing the electron fraction Y_e and weakening electron degeneracy pressure, the final source of support. Each capture also produces a neutrino that escapes, carrying away additional energy:



We encountered neutrino energy losses in Lecture 07 (pair annihilation, photoneutrinos, bremsstrahlung, plasma neutrinos) as a secondary energy sink. Here they become a primary driver of collapse — each electron capture neutrino removes both energy and a lepton from the core.

4. Gravitational instability

With no remaining pressure able to counter gravity, catastrophic collapse occurs. The inner core collapses nearly homologously (uniformly) at speeds up to $\sim 0.25c$, reaching nuclear density ($\rho \sim 2.7 \times 10^{14} \text{ g cm}^{-3}$) in less than a second. At this point the strong nuclear force resists further compression, and the core *bounces*, launching a shock wave outward.

5. Neutrino-driven explosion

The bounce shock alone stalls as it loses energy photodissociating infalling iron. The currently favored mechanism for reviving the shock is energy deposition by the intense neutrino flux streaming out of the proto-neutron star. About 3×10^{53} erg of gravitational binding energy is released — recall from Lecture 07 that $\sim GM^2/R$ sets the energy scale for self-gravitating objects, and for a neutron star ($M \sim 1.4 M_\odot$, $R \sim 10$ km) this gives $\sim 3 \times 10^{53}$ erg, about 100 times the total nuclear energy released over the star's entire main-sequence lifetime. Only $\sim 1\%$ needs to couple to the envelope to power the observed explosion ($\sim 10^{51}$ erg, or 1 "Bethe").

The result is a core-collapse supernova, leaving behind a compact remnant — usually a neutron star, sometimes a black hole. These explosions also produce many of the elements we discussed in Chapter 6, through explosive nucleosynthesis and the r-process (rapid neutron capture, introduced briefly in Lecture 07).



Periodic table showing astrophysical production sites, including core-collapse supernovae.

Pulsational Variables

A selection of intrinsically variable stars placed on an HR diagram.

Key Physical Concepts

The Instability Strip

The instability strip is a nearly vertical region on the HR diagram (roughly $T_{\text{eff}} \approx 6000\text{--}8000\text{ K}$) where stars are unstable to radial pulsations. Stars crossing this strip — whether evolving off the main sequence, on the horizontal branch, or on blue loops — can become pulsating variables. The strip spans a wide range of luminosities, so it intersects many different evolutionary stages. This is why we see Cepheids (supergiants), RR Lyrae (horizontal branch), δ Scuti (main sequence), and even ZZ Ceti (white dwarfs) all pulsating for related reasons.

The κ -Mechanism

The κ -mechanism is the primary driving mechanism for pulsations in the instability strip. Here is how it works:

1. In a normal star, compression heats the gas, which increases the opacity and makes the layer more transparent — energy leaks out easily, and the layer returns to equilibrium. This is stable.
2. However, in certain ionization zones (especially the He II $\rightarrow$ He III ionization zone at $T \approx 4 \times 10^4\text{ K}$), compression causes partial ionization rather than a temperature increase. The extra energy goes into ionizing atoms instead of heating the gas.
3. In a partially ionized zone, increasing compression *increases* the opacity (κ) — the layer becomes *more* opaque, trapping radiation. This trapped energy provides extra pressure that pushes the layer outward past equilibrium.
4. As the layer expands and cools, ions recombine, the opacity drops, trapped radiation escapes, and the layer falls back inward — overshooting again.
5. This cycle repeats, driving sustained pulsations. It acts like a heat engine: the ionization zone absorbs heat during compression and releases it during expansion, doing net work on the stellar envelope.

The instability strip has well-defined blue (hot) and red (cool) edges. On the hot side, the ionization zone lies too close to the surface (too little mass above it to drive pulsation). On the cool side, convection becomes efficient enough to transport the trapped energy before it can drive pulsation.

Pulsation Modes

Stars can pulsate in different modes, classified by the restoring force:

- Pressure (p) modes: The restoring force is pressure (compression). These are essentially sound waves trapped inside the star. They have shorter periods and higher frequencies. Radial pulsations (the entire star expanding and contracting symmetrically) are the simplest p-modes. β Cephei and δ Scuti stars exhibit p-modes.
- Gravity (g) modes: The restoring force is buoyancy (gravity acting on displaced fluid elements). These have longer periods and lower frequencies. They involve primarily horizontal displacements and are sensitive to conditions deep inside the star. γ Doradus stars and ZZ Ceti white dwarfs exhibit g-modes.
- Mixed modes: In evolved stars (e.g., red giants observed by *Kepler*), modes can behave like p-modes in the envelope and g-modes in the core. These are powerful probes of internal structure — the field of *asteroseismology*.

The Period-Luminosity Relation (Leavitt's Law)

Henrietta Leavitt discovered in 1912 that for Classical Cepheids in the Small Magellanic Cloud, brighter stars pulsate with longer periods. This period-luminosity (P-L) relation can be understood physically: more luminous Cepheids are larger, and larger stars have lower mean densities, so by the period-mean density relation ($P \propto \bar{\rho}^{-1/2}$) they pulsate more slowly.

An approximate form is:

$$M_V \approx -2.76 \log_{10} \left(\frac{P}{\text{days}} \right) - 1.4$$

By measuring a Cepheid's period (easy — just monitor its brightness over time) we get its absolute magnitude, and comparing with the apparent magnitude gives the distance. This makes Cepheids one of the most important rungs on the cosmic distance ladder. Edwin Hubble used Cepheids in the Andromeda galaxy (M31) to prove that "spiral nebulae" were external galaxies, and Cepheids remain a primary calibrator for the Hubble constant H_0 .

RR Lyrae stars serve a similar role but are fainter and have a narrower luminosity range ($M_V \approx +0.6$), making them useful distance indicators for globular clusters and the local group.

Major Classes

Classical Cepheids (Type I)

Young, metal-rich stars crossing the instability strip, spectral types F6–K2. Periods range from days to months. Pulsations are primarily radial, driven by the κ -mechanism. They follow the period-luminosity relation and are the premier extragalactic distance indicators.

W Vir Variables (Type II Cepheids)

Similar behavior to Classical Cepheids, but older, metal-poor, and lower-mass stars. They are ~1.5 mag fainter than Type I Cepheids at the same period — confusing the two types leads to systematic

distance errors (historically, this was the source of a factor-of-two error in the extragalactic distance scale).

RR Lyrae

Horizontal branch stars in the instability strip, spectral types A2–F2. Periods range from a few hours to about a day. Important as distance indicators for globular clusters and nearby galaxies. Their near-constant absolute magnitude makes them "standard candles" complementary to Cepheids.

β Cephei

Massive main-sequence (or near main-sequence) stars of spectral type B. Pulsations are dominated by pressure (p) modes, driven by the iron opacity bump (a κ -mechanism variant where iron-group elements at $T \sim 2 \times 10^5$ K provide the opacity increase). Periods are a few hours.

γ Doradus

Main-sequence stars of spectral type F. Pulsations are gravity (g) modes with longer periods (~1 day), likely driven by convective blocking at the base of the convective envelope — convection modulates the radiative flux entering from below, exciting g-modes.

δ Scuti

Stars of spectral type A to early F, on or near the main sequence. Exhibit both radial and nonradial pulsations with short periods (tens of minutes to hours). They sit at the intersection of the classical instability strip and the main sequence.

ZZ Ceti (DAV)

Hydrogen-atmosphere white dwarfs on the white dwarf cooling sequence. Pulsations are nonradial g-modes, often many simultaneously. A corresponding instability strip exists for helium-atmosphere (DB) white dwarfs. These pulsations are driven by a κ -mechanism in the hydrogen (or helium) partial ionization zone at the white dwarf surface.

Mira Variables

Highly luminous asymptotic giant branch (AGB) stars with periods ranging from ~100 to 700 days. Pulsations are predominantly radial. Their large amplitudes (several magnitudes in the visual) are due to the formation and destruction of molecules (especially TiO) in their cool atmospheres, which strongly modulates the optical opacity. Closely related are Semi-Regular Variables (SRVs).

Most stars above the main sequence and redward of the Cepheid instability strip show some level of variability.

In-Class Exercise: Pulsational Variables

Learning Goals

1. Understand different types of pulsating variables and their HR diagram placement.
2. Compute pulsation periods using the period-mean density relation.

3. Simulate simple radial pulsation light curves in Python.

Pulsation Periods

The pulsation period of a star can be approximated using the period-mean density relation:

$$P = Q \sqrt{\frac{1}{\bar{\rho}}}$$

where Q is the pulsation constant and $\bar{\rho}$ is the mean density of the star, given by

$$\bar{\rho} = \frac{3M}{4\pi R^3}$$

where M and R are in solar units.

In the following, you will compute pulsation periods using the period-mean density relation.

We will use approximate values for different pulsating stars:

Star Type	Mass ($M_{\odot}$)	Radius ($R_{\odot}$)	Q (days)
Cepheid	5	40	0.04
RR Lyrae	0.7	5	0.033
δ Scuti	2	3	0.033
Mira	1	200	0.09

We want to compute the pulsation period for each type.

► Solution

Compare Pulsation Periods

1. Plot the pulsation periods on a logarithmic scale to compare different types of variable stars.
2. Identify which stars are long-period and which are short-period variables.

► Solution

Explosive Variables

Novae

Nova and recurrent nova explosions are outbursts fueled by degenerate hydrogen ignition on the surface of a white dwarf. In Lecture 07 we studied hydrogen burning via the pp chains and CNO cycles under main-sequence conditions — steady, thermally regulated fusion. In a nova, the physics

is the same (primarily the CNO cycle, since the accreted material is CNO-enriched and the base temperatures are high), but the conditions are dramatically different: hydrogen ignites under electron-degenerate conditions, where the pressure is independent of temperature. This means the usual thermal regulation fails — a temperature increase does not cause expansion and cooling, so the burning runs away explosively. This is the thermonuclear runaway.

A drawing of a cataclysmic variable system is shown in HKT Fig. 2.25. Mass is gravitationally drawn off a donor star and forms an accretion disk around the white dwarf.



Classical novae may release a total of 10^{44} to 10^{45} ergs upon eruption, with roughly 10^{38} ergs radiated in the optical. Mass lost during the explosion ranges from 10^{-5} to 10^{-4} solar masses. The white dwarf itself survives the explosion — unlike a supernova. If accretion resumes, the process can repeat: these are recurrent novae, with observed recurrence times of decades to centuries.

The speed class of a nova — how quickly its light curve declines — correlates with the white dwarf mass. More massive white dwarfs have higher surface gravity, requiring less accreted mass to reach ignition, and the resulting explosions are faster and more luminous. This leads to the maximum magnitude–rate of decline (MMRD) relation, another distance indicator.

The Chandrasekhar Mass

A critical mass scale for understanding both novae and supernovae is the Chandrasekhar mass:

$$M_{\text{Ch}} = \frac{5.83}{\mu_e^2} M_{\odot} \approx 1.44 M_{\odot}$$

where μ_e is the mean molecular weight per electron ($\mu_e = 2$ for a C/O white dwarf). This is the maximum mass that can be supported by electron degeneracy pressure. A white dwarf that approaches M_{Ch} — either by accretion or merger — becomes unstable. For Type Ia supernovae, this instability triggers explosive thermonuclear burning. For core-collapse supernovae, the iron core exceeds M_{Ch} and collapses.

The derivation balances the electron degeneracy pressure (from the Pauli exclusion principle forcing electrons into high-momentum states) against gravity. In the relativistic limit, both the pressure and gravity scale as $\rho^{4/3}$, so there is a unique mass above which equilibrium is impossible — this is M_{Ch} .

Supernovae

Supernovae (SN or SNe) are the most spectacular stellar variables. At maximum light, they can be as bright as a small galaxy ($M_V \sim -19$ for Type Ia). Recognizing supernovae between 1900 and 1925 helped establish the size of the Milky Way and the existence of other galaxies.

Type I Supernovae

Type I spectra show no hydrogen lines, indicating the progenitor has lost its hydrogen envelope before the explosion.

Type Ia

- Occur in a wide range of galaxies and stellar populations (both young and old).
- No evidence for the formation of a neutron star or other compact remnant — the white dwarf is completely disrupted.
- The energy source is explosive fusion of about one solar mass of carbon and oxygen to iron-peak elements, especially ^{56}Ni . This is essentially the carbon and oxygen burning we studied in Lecture 07, but occurring explosively throughout the entire white dwarf on a timescale of seconds rather than the centuries-to-millennia timescales of hydrostatic burning in massive star cores. The competition between the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction and the triple-alpha process during prior helium burning (Lecture 07) determined the C/O ratio of the white dwarf, which in turn affects the ^{56}Ni yield and peak luminosity of the explosion.

Progenitor models:

The progenitor of a Type Ia supernova is a carbon-oxygen white dwarf in a binary system. Two leading scenarios exist:

1. Single-degenerate (SD): A white dwarf accretes hydrogen or helium from a non-degenerate companion star (e.g., a red giant or main-sequence star). As the WD approaches M_{Ch} , carbon ignition occurs in the interior and a thermonuclear runaway incinerates the entire star.
2. Double-degenerate (DD): Two white dwarfs in a close binary lose orbital energy via gravitational wave emission, spiral inward, and merge. If the combined mass exceeds M_{Ch} (or if the merger dynamics trigger detonation even below M_{Ch}), an explosion results.

Both channels likely contribute to the observed Type Ia rate, and disentangling them is an active area of research. Some sub-luminous Type Ia events may come from sub-Chandrasekhar-mass detonations.

The Phillips relation and dark energy:

The peak luminosity of a Type Ia SN correlates with the width of its light curve: brighter events decline more slowly. This Phillips relation (or "wider-brighter" relation) allows us to standardize Type Ia luminosities to ~7% precision in distance. Using this technique, two teams (the Supernova Cosmology Project and the High-z Supernova Search Team) discovered in 1998 that the expansion of the universe is accelerating — implying the existence of dark energy. This discovery was awarded the 2011 Nobel Prize in Physics.

 type1a_lc

A selection of light curves for Type Ia supernovae. Their uniformity (after light-curve shape correction) makes them excellent standardizable candles.

Type Ib and Ic

While not the focus of this lecture, Type Ib supernovae show helium lines but no hydrogen, and Type Ic show neither. These are core-collapse events from massive stars that lost their envelopes through stellar winds or binary interaction (Wolf-Rayet stars). Type Ic-BL (broad-lined) events are sometimes associated with long-duration gamma-ray bursts.

Type II Supernovae

Type II spectra show strong hydrogen lines and result from the collapsing cores of massive stars that retained their hydrogen envelopes.

- Almost always occur in galaxies with recent star formation, in regions with young Population I stars.
- Emit in radio and X-rays, usually after the visible peak, as the shock wave interacts with circumstellar material.
- Evidence includes the presence of a pulsar in the Crab Nebula (SN 1054) and the neutrino burst from SN 1987A.



Light curves for Type II plateau (II-P) and linear (II-L) subtypes, along with SN 1987A and SN 1993J. Radioactive ^{56}Co decay powers the late-time light curve, with a decline of roughly one magnitude per 100 days.

- Plateau subtype (II-P): After peak, the light curve remains roughly constant for ~ 100 days. This plateau is powered by hydrogen recombination: as the expanding, cooling ejecta reaches $T \approx 5500$ K, hydrogen recombines from the outside in, releasing the thermal energy that was deposited by the initial shock. The plateau ends when the recombination front reaches the base of the hydrogen envelope.
- Linear subtype (II-L): The light curve declines steadily from peak. These progenitors have thinner hydrogen envelopes, so the recombination phase is shorter and less pronounced.

SN 1987A

SN 1987A in the Large Magellanic Cloud was first observed on February 24, 1987. It was the nearest supernova since Kepler's SN in 1604 and the first to be detected in neutrinos. Two key observations:

- Neutrinos: Detected ~ 0.25 days before optical discovery by the Kamiokande-II (Japan) and IMB (US) detectors. About 20 neutrino events were recorded over ~ 10 seconds, consistent with the predicted release of $\sim 3 \times 10^{53}$ erg in neutrinos from neutron star formation. This confirmed the basic picture of core collapse.

- Gamma-ray lines of ^{56}Co : The decay sequence $^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$ both produces ^{56}Fe and powers the expanding ejecta. Recall from Lecture 07 that rapid silicon burning (where weak interactions cannot keep pace) produces ^{56}Ni rather than ^{56}Fe directly. The gamma-ray lines appeared earlier than expected, indicating that the ^{56}Ni was mixed outward into the envelope — evidence for large-scale Rayleigh-Taylor instabilities during the explosion.

The progenitor was identified as the blue supergiant Sanduleak —69° 202 (spectral type B3 I), which was surprising — models had predicted red supergiants. This led to revisions in our understanding of pre-supernova evolution, particularly the role of metallicity and mass loss.



Spectra of the four major supernova types about one week after maximum light. The ordinate is magnitude (or flux f_ν), and the abscissa is rest-frame wavelength.

Summary: From Stellar Evolution to Explosions

Massive stars ($M \gtrsim 8 M_\odot$) burn successive fuels ($\text{H} \rightarrow \text{He} \rightarrow \text{C/O} \rightarrow \text{Ne/Mg} \rightarrow \text{O/Si} \rightarrow \text{Fe}$) on progressively shorter timescales (millions of years down to days). Once an iron core forms, fusion ceases to be exothermic and collapse ensues.

Progenitor mass	Likely explosion	Typical remnant
8--10 $M_\odot$	Electron-capture SN	Neutron star
$\gtrsim 10 M_\odot$	Core-collapse (Type II) SN	Neutron star or black hole
Low-mass binary (WD accretor)	Type Ia SN or nova	None (Ia) or surviving WD (nova)

Energy Sources for Supernova Light Curves

	Early-time source	Late-time source
Type Ia	^{56}Ni decay	$^{56}\text{Co} \rightarrow ^{56}\text{Fe}$ decay
Type II	Shock heating + H recombination	$^{56}\text{Co} \rightarrow ^{56}\text{Fe}$ decay

The late-time decline rate of ~ 1 mag per 100 days in both types is set by the ^{56}Co half-life (77.3 days).

Observational Diagnostics

- Type Ia: No hydrogen; strong Si II absorption near maximum; uniform peak luminosity (standard candles).
- Type II: Strong hydrogen lines; light curves show plateau (II-P) or linear decline (II-L); broader range of peak luminosities.

The bigger picture

- The IMF determines the relative numbers of massive vs. low-mass stars, and therefore the supernova rate.
- Supernova feedback (energy, momentum, nucleosynthesis) drives the chemical evolution of galaxies and regulates subsequent star formation.

White Dwarfs, Neutron Stars, and Black Holes

These are the compact endpoints of stellar evolution. All three are supported (or not, in the case of black holes) against gravity by fundamentally different physics, and understanding them requires going beyond the classical stellar structure equations. In Lecture 07 we established that the fundamental energy scale for any self-gravitating object is $\sim GM^2/R$ — this quantity sets the gravitational binding energy, the thermal energy (via the virial theorem), the rotational energy budget, and the maximum magnetic energy. For compact remnants, this energy scale becomes enormous because R is so small.

White Dwarfs

White dwarfs are the remnants of low- and intermediate-mass stars ($M_{\text{initial}} \lesssim 8 M_{\odot}$) that have shed their envelopes. Key properties:

- Typical mass: $0.5\text{--}0.8 M_{\odot}$ (most cluster around $\sim 0.6 M_{\odot}$)
- Typical radius: $\sim 0.01 R_{\odot} \approx R_{\oplus}$ (roughly Earth-sized)
- Typical density: $\sim 10^6 \text{ g cm}^{-3}$
- Support mechanism: Electron degeneracy pressure — the Pauli exclusion principle prevents electrons from being compressed into the same quantum state. This pressure is independent of temperature, which is why white dwarfs cool at nearly constant radius.
- Composition: Mostly C/O (from He burning), though low-mass He white dwarfs and massive O/Ne/Mg white dwarfs also exist. The C/O ratio is set by the competition between the triple-alpha process and the $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ reaction during helium burning, as we discussed in Lecture 07.
- No nuclear energy source: In Lecture 07, we noted that white dwarfs are powered solely by gravitational contraction and cooling — their Kelvin-Helmholtz timescale can exceed the age of the universe. We also saw that crystallization releases latent heat ($\sim k_B T_m$ per ion), which further extends the cooling timescale.

An important consequence of degeneracy is the mass-radius relation: more massive white dwarfs are *smaller*. As mass increases, electrons become relativistic, and the maximum mass is the Chandrasekhar limit ($\approx 1.44 M_{\odot}$). Beyond this, electron degeneracy cannot support the star.

White dwarfs cool over billions of years, eventually becoming cold, dark objects. The cooling timescale depends on the mass and composition. The white dwarf luminosity function — the number of white dwarfs as a function of luminosity — provides an independent estimate of the age of the

Galactic disk ($\sim 8\text{--}10$ Gyr), because the faintest (oldest) white dwarfs have had the longest time to cool.

Neutron Stars

Neutron stars are the remnants of core-collapse supernovae from stars with $M_{\text{initial}} \sim 8\text{--}25 M_{\odot}$ (the upper bound is uncertain). Key properties:

- Typical mass: $1.2\text{--}2.0 M_{\odot}$ (the most precisely measured masses are from binary pulsars)
- Maximum mass: $\sim 2.0\text{--}2.3 M_{\odot}$, set by the nuclear equation of state (the Tolman-Oppenheimer-Volkoff limit — the neutron star analog of the Chandrasekhar mass)
- Typical radius: $\sim 10\text{--}13$ km
- Typical density: $\sim 10^{14}\text{--}10^{15} \text{ g cm}^{-3}$ (comparable to an atomic nucleus)
- Support mechanism: Neutron degeneracy pressure, nuclear forces, and potentially exotic states of matter (quark matter, hyperons). At these densities, the equation of state is uncertain and is a major area of current research.
- Gravitational binding energy: $\sim GM_{\text{NS}}^2/R_{\text{NS}} \sim 3 \times 10^{53}$ erg. As we computed in Lecture 07, this is about 100 times the total nuclear energy released during the star's entire main-sequence lifetime, and it is this energy — released as neutrinos during core collapse — that powers the supernova explosion.
- Surface gravity: $\sim 10^{11}$ times Earth's — general relativity is essential. The pressure-gravity balance is described by the Tolman-Oppenheimer-Volkoff (TOV) equation:

$$\frac{dP}{dr} = -\frac{G}{r^2} \left(\rho + \frac{P}{c^2} \right) \left(m + \frac{4\pi r^3 P}{c^2} \right) \left(1 - \frac{2Gm}{rc^2} \right)^{-1}$$

Each of the three correction terms in parentheses makes gravity stronger than the Newtonian prediction. The TOV equation replaces the familiar hydrostatic equilibrium equation from Newtonian stellar structure.

Observed manifestations:

- Pulsars: Rapidly rotating, highly magnetized neutron stars that emit beamed radio (and sometimes X-ray/gamma-ray) radiation. Rotation periods range from milliseconds to seconds. The Crab pulsar ($P = 33$ ms) is the remnant of SN 1054. In Lecture 07 we noted that rotational energy is also bounded by $\sim GM^2/R$; for a millisecond pulsar, the rotational energy is $\sim 10^{52}$ erg, powering a bright pulsar wind nebula.
- Magnetars: Neutron stars with extreme magnetic fields ($B \sim 10^{14}\text{--}10^{15}$ G), powered by magnetic field decay rather than rotation. They produce soft gamma repeaters (SGRs) and anomalous X-ray pulsars (AXPs). Their magnetic energy reservoir is also $\lesssim GM^2/R$.
- X-ray binaries: Neutron stars accreting from a companion star, producing X-ray emission from the hot accretion flow.

Gravitational wave constraints: The binary neutron star merger GW170817 (detected by LIGO/Virgo in 2017) provided direct constraints on the neutron star equation of state through the tidal

deformability of the merging stars. The associated electromagnetic counterpart (kilonova) confirmed that neutron star mergers are a major site of r-process nucleosynthesis — the rapid neutron capture process we introduced in Lecture 07.

Black Holes

When a stellar core exceeds the TOV limit, no known force can prevent complete gravitational collapse. The result is a black hole.

- Fully described by three quantities: mass M , angular momentum J (spin), and electric charge Q (the "no-hair theorem"). In practice, astrophysical black holes are expected to have $Q \approx 0$.
- Event horizon radius (Schwarzschild radius, for a non-spinning black hole):

$$R_S = \frac{2GM}{c^2} \approx 3.0 \text{ km} \left(\frac{M}{M_\odot} \right)$$

A $10 M_\odot$ black hole has an event horizon of only ~ 30 km.

- Stellar-mass black holes: $\sim 3\text{--}100 M_\odot$, formed from core collapse of the most massive stars or from mergers.

The mass gap: Observations suggested a gap in compact object masses between the heaviest neutron stars ($\sim 2 M_\odot$) and the lightest black holes ($\sim 5 M_\odot$). Whether this gap is real or an observational selection effect is debated. Recent gravitational wave detections (e.g., GW190814, with a $\sim 2.6 M_\odot$ secondary) and electromagnetic observations are beginning to populate this gap.

How we detect black holes: Since black holes emit no light, we infer their presence from:

- Orbital dynamics of companion stars (X-ray binaries, radial velocity surveys)
- Accretion disk emission (X-rays from hot gas spiraling inward)
- Gravitational waves from merging compact objects (LIGO/Virgo/KAGRA)
- Gravitational lensing and astrometric wobbles



In-Class Exercise: Simulating Nova Light Curves

Goal:

Explore the rise and fall of classical novae using a simple Python model. We will simulate how the optical brightness changes over time and see how different physical parameters affect the light curve.

Background:

- Classical novae occur when hydrogen accreted on a white dwarf ignites explosively.

- The brightness rises rapidly as energy is released, then declines as the ejecta expands and cools.
- We can model the light curve with a simple exponential rise and decay:

$$L(t) = L_{\text{peak}} \times \exp\left(-\frac{|t - t_{\text{peak}}|}{\tau}\right)$$

where:

- L_{peak} is the peak luminosity,
- t_{peak} is the time of maximum light,
- τ is a characteristic timescale (rise or fall).

Tasks:

1. Define parameters for a nova: L_{peak} , t_{peak} , τ .
2. Create a time array covering before and after the peak.
3. Compute $L(t)$ using the exponential model.
4. Plot $L(t)$ vs time.
5. Convert luminosity to apparent magnitude:

$$m(t) = -2.5 \log_{10}\left(\frac{L(t)}{L_0}\right)$$

Ideas to consider:

- Compare "fast" vs "slow" novae by changing τ .
- Explore how varying L_{peak} affects apparent brightness.
- Simulate observational noise.

► Solution

In-Class Exercise: Recurrent Nova Light Curves

Goal:

Extend the previous nova simulation to explore multiple eruptions in a single system. We will see how the light curve builds over time with recurrent outbursts.

Tasks:

1. Define the number of eruptions (`num_eruptions`) and the time interval between eruptions (`eruption_interval`).
2. Loop over eruptions to compute the total luminosity at each time.
3. Plot the resulting recurrent nova light curve, both in luminosity and optionally in apparent magnitude.
4. Optional: add observational noise or change rise/decay timescales for each eruption.

Extensions / Discussion:

- Compare systems with frequent vs. rare eruptions.
- Explore how changing L_{peak} , τ_{rise} , or τ_{fall} affects the overall brightness evolution.
- Discuss the physical interpretation: accretion buildup, ignition threshold, and cumulative observational effects.

► Solution

Star Formation

We have now followed stars from birth to death — tracing their energy sources (Lecture 07) and their explosive endpoints (this lecture). To complete the cycle, we return to the beginning: how do stars form in the first place?

In Lecture 07 we saw that protostars are powered by gravitational contraction on roughly the Kelvin-Helmholtz timescale, and that the distinction between stars, brown dwarfs, and gas giants depends on whether central temperatures become high enough for sustained hydrogen fusion via the pp chain. We also saw that the free-fall timescale $t_{\text{ff}} \sim \sqrt{R^3/(GM)}$ sets the fastest possible collapse rate. Here we develop these ideas further.

Within the Milky Way today, most star formation occurs in clouds of gas that are:

- Molecular clouds: mostly H_2 , with CO as an important tracer (H_2 has no permanent dipole moment, so it does not radiate efficiently at low temperatures; CO's rotational transitions at ~ 2.6 mm are the standard observational proxy)
- Cool: temperatures of 5–15 K
- Dense: 10^3 or more H_2 molecules per cm^3 , which is much thinner than air on Earth ($\sim 10^{19} \text{ cm}^{-3}$) but dense compared to the galactic average of about 1 hydrogen atom per cm^3
- Large: sizes of parsecs to tens of parsecs, and masses up to $10^5 M_{\odot}$ (giant molecular clouds, or GMCs)
- Primarily located in the spiral arms of the galactic disk, where gas is compressed by density waves

The chemical composition of these clouds — their metallicity, dust content, and molecular abundances — reflects the cumulative nucleosynthesis from previous generations of stars. The heavy elements produced in the burning stages of Lecture 07 and the supernova explosions of this lecture are recycled into new molecular clouds, enriching the next generation. This is galactic chemical evolution.

The Jeans Criterion

A cloud will collapse under its own gravity if the gravitational energy exceeds the thermal (kinetic) energy. This defines a critical mass, the Jeans mass:

$$M_J = \left(\frac{5k_B T}{G \mu m_H} \right)^{3/2} \left(\frac{3}{4\pi \rho} \right)^{1/2}$$

where T is the temperature, μ is the mean molecular weight, m_H is the hydrogen mass, and ρ is the density. A cloud with $M > M_J$ is gravitationally unstable and will begin to collapse.

Equivalently, there is a Jeans length:

$$\lambda_J = \left(\frac{\pi k_B T}{G \mu m_H \rho} \right)^{1/2}$$

Regions larger than λ_J are unstable. For a typical molecular cloud ($T \sim 10$ K, $n \sim 10^3$ cm⁻³, $\mu \approx 2.3$), the Jeans mass is $M_J \sim 10\text{--}50 M_\odot$ and the Jeans length is $\lambda_J \sim 1$ pc. Since GMCs are much more massive than M_J , they are highly Jeans-unstable and fragment into many smaller clumps, each of which can form a star or small group of stars. This fragmentation is a key reason why stars form in clusters rather than individually.

Free-Fall Time

Once a cloud (or a fragment) becomes Jeans-unstable, it collapses on roughly the free-fall timescale:

$$t_{\text{ff}} = \sqrt{\frac{3\pi}{32 G \rho}} \approx 4.4 \times 10^7 \left(\frac{n}{1 \text{ cm}^{-3}} \right)^{-1/2} \text{ yr}$$

For a molecular cloud core with $n \sim 10^4$ cm⁻³, $t_{\text{ff}} \sim 4 \times 10^5$ yr — a few hundred thousand years. Note that the free-fall time depends only on the density, not on the mass or size of the cloud.

In practice, star formation is much slower than free fall. The star formation efficiency of molecular clouds is only a few percent per free-fall time. Magnetic fields, turbulence, and feedback from newly formed stars all act to slow the collapse and regulate the rate of star formation.

From Collapse to Star

The basic sequence from cloud to star is:

1. Molecular cloud fragmentation: The cloud fragments into dense cores ($n \sim 10^4\text{--}10^5$ cm⁻³, $T \sim 10$ K, sizes ~ 0.1 pc).
2. Isothermal collapse: The core collapses nearly isothermally (dust radiation efficiently cools the gas) until the central density becomes high enough for the gas to become optically thick ($n \sim 10^{10}$ cm⁻³). At this point a first hydrostatic core forms.
3. Protostar formation: As the core heats up, H₂ dissociates at ~ 2000 K, absorbing energy and triggering a second collapse. A true protostar forms at the center.

4. Accretion phase: The protostar accretes from the surrounding envelope and disk. During this phase it is heavily obscured and visible only at infrared and (sub)millimeter wavelengths. These are Class 0 and Class I protostars. The energy source during this phase is gravitational contraction — exactly the Kelvin-Helmholtz mechanism from Lecture 07.
5. Pre-main-sequence evolution: Once the envelope disperses, the star becomes optically visible as a T Tauri star (low mass) or Herbig Ae/Be star (intermediate mass). It contracts along the Hayashi track (fully convective, nearly vertical on the HR diagram) and then along the Henyey track (developing a radiative core, moving leftward toward the main sequence).
6. Main sequence: Hydrogen ignition in the core begins — the pp chain or CNO cycle from Lecture 07 takes over from gravity as the energy source — and the star reaches the zero-age main sequence (ZAMS). The star's mass determines which burning stages it will eventually undergo (Lecture 07) and what kind of remnant it will leave behind (this lecture).

The Initial Mass Function (IMF)

A key question in star formation is how the IMF behaves at the low-mass end. How many very low-mass stars or brown dwarfs exist, whose lifetimes exceed the age of the universe? And is the IMF the same in all environments?

The IMF is often expressed as:

$$\xi(M) dM = \xi_0 M^{-\alpha} dM$$

where:

- M is the stellar mass in solar masses
- α is the IMF slope (Salpeter 1955: $\alpha \approx 2.35$)
- ξ_0 is a normalization constant

This function gives the number of stars per unit mass interval. The Salpeter slope of $\alpha = 2.35$ means that for every factor-of-10 increase in mass, there are ~ 20 times fewer stars. This steep slope has profound consequences: most of the *number* of stars are low-mass, but most of the *luminosity* and *nucleosynthetic output* come from rare, massive stars. Combined with the steep temperature dependence of nuclear burning rates from Lecture 07 ($\epsilon \propto T^\nu$ with $\nu \sim 4$ for pp, ~ 18 for CNO, ~ 40 for triple-alpha), this means that a small number of massive stars dominate the energy budget, chemical enrichment, and feedback in a galaxy.

Modern refinements:

The Salpeter power law works well for $M \gtrsim 1 M_\odot$, but observations show that the IMF flattens and turns over below $\sim 0.5 M_\odot$. Two widely used parameterizations that capture this behavior:

- Kroupa (2001): A broken power law with $\alpha = 0.3$ for $0.01\text{--}0.08 M_\odot$, $\alpha = 1.3$ for $0.08\text{--}0.5 M_\odot$, and $\alpha = 2.3$ for $M > 0.5 M_\odot$.

- Chabrier (2003): A log-normal distribution below $\sim 1 M_{\odot}$ (peaking near $0.2\text{--}0.3 M_{\odot}$) joined to a Salpeter power law above $1 M_{\odot}$.

The peak of the IMF near $0.2\text{--}0.3 M_{\odot}$ may reflect the characteristic Jeans mass in star-forming regions. Whether the IMF is truly universal or varies with environment (metallicity, density, radiation field) is one of the major open questions in astrophysics. Evidence for IMF variations exists in extreme environments (e.g., the Galactic center, massive elliptical galaxies), but the IMF appears to be remarkably consistent across normal star-forming regions in the Milky Way and nearby galaxies.

In-Class Exercise: Star Formation and the Initial Mass Function

Part 1: Cloud Density Comparison

The average density of the Milky Way is roughly 1 H atom/cm^3 . Compare the number density of a typical molecular cloud to the galactic average. Fill in the missing calculation below:

► Solution

Part 2: Number of Stars in a Mass Range

1. Consider a cloud forming stars in the mass range $0.1\text{--}50 M_{\odot}$.
2. Compute the relative number of stars in two mass ranges:
 - Low-mass: $0.1\text{--}1 M_{\odot}$
 - High-mass: $8\text{--}50 M_{\odot}$

You can do this by integrating ($M^{-\alpha}$) over the mass range.

► Solution

Part 3: IMF Visualization

Plot the IMF over a wide mass range ($0.1\text{--}50 M_{\odot}$) on a log-log scale.

- x-axis: stellar mass ($M_{\odot}$)
- y-axis: number of stars per unit mass (unnormalized)

► Solution

Supernova Classification and Light Curve Modeling

The sections above described the observational signatures of different supernova types. Next, we will build a simple quantitative model for the Type Ia light curve powered by radioactive decay. In

Lecture 07 we learned that rapid silicon burning (where the weak interaction cannot keep pace) produces ^{56}Ni rather than ^{56}Fe directly. This is true in both core-collapse and Type Ia supernovae. The subsequent radioactive decay chain $^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$ is the dominant energy source for supernova light curves after the first few days, and the exercise below models this quantitatively.

In-Class Exercise: Supernova Light Curves from Radioactive Decay

A simple model of a supernova lightcurve is just to add up the energy of the photons that result from the decay:



The evolution equations for the number densities appear as:

$$\begin{aligned}\frac{dn_{\text{Ni}}}{dt} &= -\lambda_{\text{Ni}}n_{\text{Ni}} \\ \frac{dn_{\text{Co}}}{dt} &= +\lambda_{\text{Ni}}n_{\text{Ni}} - \lambda_{\text{Co}}n_{\text{Co}}\end{aligned}$$

where λ_{Ni} is the decay constant for ^{56}Ni and λ_{Co} is the decay constant for ^{56}Co .

The initial conditions are:

$$\begin{aligned}n_{\text{Ni}}(t=0) &= n_0 \\ n_{\text{Co}}(t=0) &= 0\end{aligned}$$

We can integrate the Ni equation directly, giving:

$$n_{\text{Ni}} = n_0 e^{-\lambda_{\text{Ni}}t}$$

We define the dimensionless abundance fractions $X_{\text{Ni}} \equiv n_{\text{Ni}}/n_0$ and $X_{\text{Co}} \equiv n_{\text{Co}}/n_0$, so:

$$X_{\text{Ni}}(t) = e^{-\lambda_{\text{Ni}}t}$$

The Co equation becomes:

$$\frac{dn_{\text{Co}}}{dt} = \lambda_{\text{Ni}}n_0 e^{-\lambda_{\text{Ni}}t} - \lambda_{\text{Co}}n_{\text{Co}}$$

This can be integrated by applying an integrating factor:

$$\frac{d}{dt} (e^{\lambda_{\text{Co}}t} n_{\text{Co}}) = e^{\lambda_{\text{Co}}t} \frac{dn_{\text{Co}}}{dt} + \lambda_{\text{Co}} e^{\lambda_{\text{Co}}t} n_{\text{Co}} = e^{\lambda_{\text{Co}}t} \left[\frac{dn_{\text{Co}}}{dt} + \lambda_{\text{Co}} n_{\text{Co}} \right]$$

Now the term in the $[\dots]$ is just $\lambda_{\text{Ni}}n_{\text{Ni}}$, so this becomes:

$$\frac{d}{dt} (e^{\lambda_{\text{Co}}t} n_{\text{Co}}) = e^{\lambda_{\text{Co}}t} \lambda_{\text{Ni}} n_0 e^{-\lambda_{\text{Ni}}t}$$

This gives:

$$X_{\text{Co}}(t) = \frac{\lambda_{\text{Ni}}}{\lambda_{\text{Co}} - \lambda_{\text{Ni}}} (e^{-\lambda_{\text{Ni}}t} - e^{-\lambda_{\text{Co}}t})$$

The luminosity at time t is the rate of energy release from both decay chains. If we define $\tilde{Q}_{\text{Ni}}$ and $\tilde{Q}_{\text{Co}}$ as the total decay energy per solar mass of the parent isotope (in erg $M_{\odot}^{-1}$), and M_{Ni} is the initial mass of ^{56}Ni synthesized, then:

$$L(t) = M_{\text{Ni}} \left[\tilde{Q}_{\text{Ni}} \lambda_{\text{Ni}} X_{\text{Ni}}(t) + \tilde{Q}_{\text{Co}} \lambda_{\text{Co}} X_{\text{Co}}(t) \right]$$

Here λX gives the instantaneous decay rate (fraction per second), and multiplying by $\tilde{Q}$ and M_{Ni} converts this to a luminosity in erg s^{-1} .

Part 1: Compute Ni and Co Abundance Fractions

Using the results derived above:

$$X_{\text{Ni}}(t) = e^{-\lambda_{\text{Ni}}t}, \quad X_{\text{Co}}(t) = \frac{\lambda_{\text{Ni}}}{\lambda_{\text{Co}} - \lambda_{\text{Ni}}} (e^{-\lambda_{\text{Ni}}t} - e^{-\lambda_{\text{Co}}t})$$

► Solution

Part 2: Compute the Luminosity

$$L(t) = M_{\text{Ni}} \left[\tilde{Q}_{\text{Ni}} \lambda_{\text{Ni}} X_{\text{Ni}}(t) + \tilde{Q}_{\text{Co}} \lambda_{\text{Co}} X_{\text{Co}}(t) \right]$$

We will now compute and plot this light curve.

► Solution

Part 3: Things to try

- How does the light curve change if the initial Ni mass is increased to $0.8 M_{\odot}$?
- How does the light curve change if the Co half-life is shorter?
- Can you identify roughly when the Co decay begins to dominate the luminosity?